Most Asked JAVA Coding Programs (Automation QA/SDET)

Prepared by

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Most Asked SDET JAVA Programs

1.) Java program to Find Odd or Even number

```
import java.util.Scanner;

public class OddEven {
       public static void main(String[] args) {

       Scanner scanner = new Scanner(System.in);
       System.out.print("Enter any number: ");
       int number = scanner.nextInt();

       if (number % 2 == 0)
           { System.out.println(number + " is even.");
       } else {
           System.out.println(number + " is odd.");
       }
    }
}
```

2.) Java program to find Prime number

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A. import java.util.Scanner;

```
public class PrimeNumber {

       public static void main(String[] args)
      {      Scanner      scanner      =      new
      Scanner(System.in);
      System.out.print("Enter a number: ");
      int number = scanner.nextInt();

      if (isPrime(number)) {
         System.out.println(number + " is a prime number.");
       } else {
         System.out.println(number + " is not a prime number.");
       }
    }

public static boolean isPrime(int num)
      { for (int i = 2; i <= num / 2; i++)
      {
       //try each number by using %
         if (num % i == 0)
           { return false;
     }
  }          return true;
  }
```

3.) Java program to find Fibonacci series upto a given number range

B. `import java.util.Scanner;`

```
public class PrimeNumber {

    public static void main(String[] args)
       { Scanner sc = new Scanner(System.in);
       System.out.println("enter number of terms");
       int number = 6;
       int first = 0, second = 1, next;
       System.out.println("Fibonacci series is ");
       for ( int i = 0; i<=number; i++)
         {
             System.out.println(first + "");
             next = second+first;
             first = second;
             second = next;
         }
}
```

Output: 0 1 1 2 3 5 8

4.) Java program to swap two numbers without using third variable

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```
import java.util.Scanner; public class

SwapNumbers {
     public static void main(String[] args) {

   Scanner scanner = new Scanner(System.in);
   System.out.print("Enter the first number: ");
   int a = 5,
   System.out.print("Enter the second number: ");
   int b = 10;
   System.out.println("Before swapping: a = " + a + ", b = " + b);

   a = a + b;
   b = a - b;
   a = a - b;
   System.out.println("After swapping: a = " + a + ", b = " + b);

    }
}
```

```
Output: After Swapping: a = 10 , b = 5
```

5.) Java program to Find Factorial on given Number

```
import java.util.Scanner;

public class FactorialNumber {

    public static void main(String[] args)
    { int factorial =1;
    Scanner scanner = new Scanner(System.in);
    System.out.print("Enter any number ");
    int number = 5;

    for (int i = 1; i <= number;
     i++){ factorial = factorial * i;
    }
    System.out.println("Factorial number is :" +factorial);

    }
}
```

```
Input: 5!
```

Output: 5! = 5*4*3*2*1 = 120

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6.) Java program to Reverse Number

```
import java.util.Scanner;

public class ReverseNumber {

   public static void main(String[] args)
    { int no, rev=0,r,a;
    Scanner scanner = new Scanner(System.in);
    System.out.println("Enter any number : ");
    no = scanner.nextInt();
    a = no;
    while(no>0)
    {
        r =  no%10; rev
        = rev*10+r;
        no=no/10;
    }

    System.out.println("Reverse : " +rev);

    }
}
```

```
Input: 15786
Output: 68751
```

7.) Java program to find Armstrong Number

C. import java.util.Scanner;

```
public class ArmstrongNumber {

    public static void main(String[] args)
    { int arm=0, a,b,c,d,no;
    Scanner scanner = new Scanner(System.in);
    System.out.println("Enter any number : ");
    no = scanner.nextInt();
    d = no;
    while(no>0)
    {
        a =  no%10;
        no = no/10;
        arm =arm+a*a*a;
    }
    if(arm==d){ System.out.println("Armstro
    ng number");
    }
    else{
    System.out.println("Not Armstrong number");
    }
}
}
```

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8.) Java program to find number of digits in given number

```
import java.util.Scanner;
public class NumberOfDigits {

  public static void main(String[] args)
{ int no = 0, a = 0;
Scanner scanner = new Scanner(System.in);
System.out.println("Enter any number : ");
no = scanner.nextInt();
if(no<0)
{
    no = no * -1;

} else if (no==0)
    { no=1;
}
while(no>0)
{
    no=no/10;
    a++;}
System.out.println("Number of digits in given number is :" +a); }
```

9.) Java program to find Palindrome number

```
import java.util.Scanner;

 public class Main {
    public static void main(String[] args)
        { Scanner scanner = new
        Scanner(System.in);
         System.out.print("Enter a number: ");
         int number = scanner.nextInt();

         if (isPalindrome(number))
          { System.out.println(number + " is a
          palindrome.");
       } else {
          System.out.println(number + " is not a palindrome.");
       }
  }


  public static boolean isPalindrome(int num)
      { int originalNumber = num;
      int reversedNumber = 0;

      while (num != 0) {
          int digit = num % 10;
          reversedNumber = reversedNumber * 10 + digit;
          num =  num/10;
      }

      return originalNumber == reversedNumber;
  }
}
```

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```
Enter a number: 1001

1001 is a palindrome.
```

10.) Java program to calculate the sum of digits of a number

```
public class Main {
    public static void main(String[] args)
        { int number = 12345;

        int sumOfDigits = calculateSumOfDigits(number);

        System.out.println("Sum of digits of " + number + " is: " +
sumOfDigits);
    }

    public static int calculateSumOfDigits(int number)
        { int sum = 0;
        while (number > 0) {
            int digit = number % 10; // Extract the last digit
            sum = sum + digit; // Add the digit to sum
            number = number / 10; // Remove the last digit from number
        }
        return sum;
    }
}
```

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II. Output:

Sum of digits of 12345 is: 15

Strings

1.) Java program to reverse a string

```
import java.util.Scanner;
 public class Test {
     public static void main(String[] args)
       { Scanner scanner = new
       Scanner(System.in);
       System.out.print("Enter a string: ");
       String input = scanner.nextLine();
       char ch;
       String nstr = "";
       for (int i = 0; i < input.length(); i++)
            { ch = input.charAt(i);
            nstr = ch + nstr;
         }
 System.out.println("Reversed String is : " + nstr);
```

2.) Java program to reverse each word of a given string

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```
public static void main(String[] args) {
    reverseEachWordOfString("Java is good programming langauges");
}

static void reverseEachWordOfString(String inputString)
{
    String[] words = inputString.split(" ");

    String reverseString = "";
    for (int i = 0; i < words.length; i++)
            { String word = words[i];
            String nstr = "";
            char ch;
            for (int j = 0; j < word.length(); j++)
                   { ch = word.charAt(j);
                   nstr = ch + nstr;
    }
    reverseString = reverseString + nstr + " ";
}

    System.out.println(inputString);
    System.out.println(reverseString);
}
```

```
Input:  Java is good programming langauges
Output: avaJ si doog gnimmargorp seguagnal
```

3.) Java program to find duplicate characters in a string

```
import java.util.HashMap;
import java.util.Set;

public class Main {

public static void main(String[] args) {
        duplicateCharacterCount("Learn Java Programming");
}

static void duplicateCharacterCount(String inputString) {

    HashMap<Character, Integer> charCountMap = new HashMap<>();
    char[] strArray = inputString.toCharArray();
    for (char c : strArray) {
        if (charCountMap.containsKey(c))
            { charCountMap.put(c, charCountMap.get(c) +
            1);
        } else {
            charCountMap.put(c, 1);
        }
    }

    Set<Character> charsInString = charCountMap.keySet();
    System.out.println("Duplicate Characters in : " + inputString);

    for (Character ch : charsInString)
        { if (charCountMap.get(ch) > 1) {
            System.out.println(ch + " : " + charCountMap.get(ch));
        }
    }
}
}
```

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```
Duplicate Characters in : Learn Java Programming

a : 4

g : 2

m : 2

n : 2

r : 3
```

4.) Java program to count Occurrences of Each Character in String

```
import java.util.HashMap;

public class Main {

    public static void main(String[] args) {
        CharacterCount("Test Automation Java Automation");
    }

    static void CharacterCount(String inputString) { HashMap<String,Integer>
      charCountMap = new HashMap<>(); for(String s : inputString.split(" "))
      {
          if(charCountMap.containsKey(s))
          {
              charCountMap.put(s,charCountMap.get(s)+1);
          }
          else
          {
              charCountMap.put(s,1);
          }
      }
        System.out.println("Count of Characters in a given string : " +
charCountMap);
          }
      }
```

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Count of Characters in a given string : **{Java=1, Automation=2, Test=1}**

5.) Java program to count the number of words in a string

```
public class Main {
   public static void main(String[] args)
  { System.out.println("Enter the String");
  Scanner sc = new Scanner(System.in);
  String s = sc.nextLine();
  int count = 1;

  for (int i = 0; i < s.length() - 1; i++) {
      if ((s.charAt(i) == ' ') && (s.charAt(i + 1) != ' '))
          { count++;
      }
  }
  System.out.println("Number of words in a string: " +count);    }
}
```

```
Enter the String:  Welcome to Java World
Number of words in a string:  4
```

6.) Java program to find all permutations of a given string

```
import java.util.Scanner;

 public class Main {
    public static void main(String[] args)
       { String str = "abc";
       permute(str, "");
    }

    static void permute(String str, String prefix)
        { if (str.length() == 0) {
            System.out.println(prefix);
        } else {
            for (int i = 0; i < str.length(); i++) {
                String rem = str.substring(0,i) + str.substring(i+1);
                permute(rem,prefix + str.charAt(i));
            }
        }
    }
}
```

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```
abc

acb

bac

bca

cab

cba
```

7.) Java program to find if a string is Palindrome

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args)
        { String str = "madam";
        System.out.println(isPalindrome(str));
    }

    static boolean isPalindrome(String str) { int start = 0;
        int end = str.length() - 1;

        while (start < end) {
            if (str.charAt(start) != str.charAt(end))
                { return false;
            }
            start++;
            end--;
        }
        return true;
    }
}
```

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8.) Java program to determine if Two Strings are Anagrams

```
public class Main {

    public static void main(String[] args)
        { String str1 = "listen";
        String str2 = "silent"; System.out.println(areAnagrams(str1,str2));
    }

    static boolean areAnagrams(String str1, String str2)
        { if(str1.length() != str2.length())
        {
            return false;
        }

        int[] charCount = new int[256];
        for( int i = 0; i < str1.length(); i++)
        {
            charCount[str1.charAt(i)]++;
            charCount[str2.charAt(i)]--;
        }

        for ( int count : charCount)
        {
            if ( count !=0 )
            {
                return false;
            }
        }
        return true;
    }
}
```

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9.) Java program to Count Vowels and Consonants in a given string

```
public class Main {
    public static void main(String[] args) { String str = "Hello
            World"; VowelConsonantCount(str);
}

static void VowelConsonantCount(String str)
    { int vowels = 0, consonants = 0;
    str = str.toLowerCase();
    for (char c : str.toCharArray())
        { if (c >= 'a' && c <= 'z') {
            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u')
{
                vowels++;
            } else {
                consonants++;
            }
        }
    }
    System.out.println("Vowels : " + vowels);
    System.out.println("Consonants : " + consonants);
}
}
```

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Vowels : 3

Consonants : 7

10.) Java program to print unqiue characters

```
import java.util.Scanner;

public class Main {
   public static void main(String[] args)
       { Scanner scanner = new
       Scanner(System.in);
       System.out.print("Enter a string: ");
       String input = scanner.nextLine();

       System.out.println("Unique characters in \"" + input + "\":");
       printUniqueCharacters(input);

}
```

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```
public static void printUniqueCharacters(String str) {
    // Assume ASCII characters (0-127), use boolean array to track character occurrences
    boolean[] unique = new boolean[128];

    for (int i = 0; i < str.length(); i++)
        { char ch = str.charAt(i);
        if (!unique[ch])
            { unique[ch] = true;
            System.out.print(ch + " ");
        }
    }

}
}
```

Enter a string: **Java Automation**

```
Unique characters in "Java Automation":

J a v  A u t o m i n
```

11.) Java program to print even indexed characters

```
import java.util.Scanner;

public class Main {
public static void main(String[] args)
      { Scanner scanner = new
      Scanner(System.in);
      System.out.print("Enter a string: ");
      String input = scanner.nextLine();

  System.out.println("Even indexed characters in \"" + input + "\":");
printEvenIndexedCharacters(input);

}


public static void printEvenIndexedCharacters(String str) { for (int i = 0; i <
      str.length(); i++) {
          if (i % 2 == 0) {
              System.out.print(str.charAt(i));
          }
      }

}
}
```

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```
Enter a string: Automation

Even indexed characters in "Automation":
```

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12.) Java program to remove space from a given string

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args)
        { Scanner scanner = new
        Scanner(System.in);
        System.out.print("Enter a string with spaces: ");
        String input = scanner.nextLine();

        String stringWithoutSpaces = removeSpaces(input); System.out.println("String
        without spaces: " +
stringWithoutSpaces);
    }

    public static String removeSpaces(String str)
        {    StringBuilder    result    =    new
        StringBuilder();  for  (int  i  =  0;  i  <
        str.length(); i++) {
            if   (str.charAt(i)   !=   '   ')
                { result.append(str.charAt(i)
                );
            }
        }
        return result.toString();
    }
}
```

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```
Enter a string with spaces: Welcome to Java World
String without spaces: WelcometoJavaWorld
```

13.) Java program to print each letter twice from a given string

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = scanner.nextLine();

        String doubledString = doubleCharacters(input);
        System.out.println("Doubled characters: " + doubledString);
    }

    public static String doubleCharacters(String str) {

        StringBuilder doubled = new StringBuilder();
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i); doubled.append(ch).append(ch); // Append each
            character
 twice

        }
        return doubled.toString();
    }
}
```

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```
Enter a string: hello
Doubled characters: hheellllloo
```

14.) Java program to swap two string without using 3rd variable

```
import java.util.Scanner;

public class Main {
   public static void main(String[] args)
         { Scanner scanner = new
         Scanner(System.in);
         System.out.print("Enter first string: ");
         String str1 = scanner.nextLine();
         System.out.print("Enter second string: ");
         String str2 = scanner.nextLine();

         System.out.println("Before swapping: str1 = " + str1 + ",
str2 = " + str2);

         // Swapping without using a third variable
         str1 = str1 + str2; // Concatenate str1 and str2 and store in str1
         str2 = str1.substring(0, str1.length() - str2.length());
// Extract the initial part (original str1) from the concatenated string
         str1 = str1.substring(str2.length()); // Extract the remaining part (original str2)
from the concatenated string

         System.out.println("After swapping: str1 = " + str1 + ",
str2 = " + str2);
    }
}
```

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```
Enter first string: Hello

Enter second string: World
```

Before swapping: str1 = **Hello**, str2 = **World**

After swapping: str1 = **World**, str2 = **Hello**

15.) Java program to gives Output: a2b2c3d2 for the Input String Str = “aabbcccdd”

```
import java.util.Scanner;

public class Main {
   public static void main(String[] args)
        { Scanner scanner = new
        Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = scanner.nextLine();

        String output = getCharacterCount(input); System.out.println("Output: " + output);
   }


   public static String getCharacterCount(String str)
        { StringBuilder result = new StringBuilder();
        int count = 1;

        for (int i = 0; i < str.length(); i++) {
            // If the next character is the same, increase the count
            if (i + 1 < str.length() && str.charAt(i) == str.charAt(i
+ 1)) {
                count++;
            } else {
                // Append the character and its count to the result
                result.append(str.charAt(i)).append(count); count = 1; //
                Reset the count
            }
        }

        return result.toString();
   }
}
```

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```
Enter a string: aabbcccdd

Output: a2b2c3d2
```

16.) Java program to gives two Output: “abcde”, “ABCDE” for the Input String Str = “aBACbcEDed”

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args)
       { Scanner scanner = new
       Scanner(System.in);
       System.out.print("Enter a string: ");
       String input = scanner.nextLine();
                   System.out.println("Original String is: "+ input);
                   separateCharacters(input);
        }

     public static void separateCharacters(String input)
        {
         StringBuilder lowerCase = new StringBuilder();
         StringBuilder upperCase = new StringBuilder();

         for(char ch : input.toCharArray())
         {
              if(Character.isLowerCase(ch))
              {
                   lowerCase.append(ch);
              }
              else
              {
                   upperCase.append(ch);
              }
         }
         System.out.println("Output in lowercase: "+lowerCase);
         System.out.println("Output in uppercase "+upperCase);
      }
```

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```
Enter a string: aBACbcEDed

Output in lowercase: abced

Output in uppercase: ABCED
```

17.) Java program to gives two Output: “Subburaj”, “123” for the Input String Str = “Subbu123raj”

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args)
       { Scanner scanner = new
       Scanner(System.in);
       System.out.print("Enter a string: ");
       String input = scanner.nextLine();
                   System.out.println("Original String is: "+ input);
                   separateAplhaAndNumeric(input);
        }

     public static void separateAlphaAndNumeric(String input)
        {
         StringBuilder alphaPart = new StringBuilder();
         StringBuilder numericPart = new StringBuilder();

         for(char ch : input.toCharArray())
         {
              if(Character.isLetter(ch))
              {
                   alphaPart.append(ch);
              }
              else if (Character.isDigit(ch))
              {
                   numericPart.append(ch);
              }
         }

         System.out.println("Output in Alpha: "+alphaPart.toString());
         System.out.println("Output in Numeric:
"+numericPart.toString());
      }
```

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```
Enter a string: Subbu123raj

Output in lowercase: Subburaj

Output in uppercase: 123
```

18.) Java program to gives Output: "32412120000" for the Input String Str = "32400121200"

```
 public class Main {
    public static void main(String[] args)
        { String input = "32400121200";
        String output = rearrangeDigits(input);
        System.out.println("Output: " + output);
}

public static String rearrangeDigits(String input) {
    // Split the input into parts: digits and non-digits StringBuilder digits =
        new StringBuilder(); StringBuilder nonDigits = new StringBuilder();

        for (char c : input.toCharArray())
            { if (Character.isDigit(c)) {
             digits.append(c);
           } else
             { nonDigits.append(c);
         }
    }

    // Concatenate non-digits followed by digits
      return digits.toString() + nonDigits.toString();
}
}
```

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```
Output: 32412120000
```

19.) Java program to gives Output: "00003241212" for the Input String Str = "32400121200"

```
public class Main {
    public static void main(String[] args)
         { String input = "32400121200";
         String formattedOutput = String.format("%011d", Long.parseLong(input));
    System.out.println("Formatted output: " + formattedOutput);
}
}
```

```
Formatted output: 00003241212
```

20.) Java program to find the longest without repeating characters

```
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {
        String s1 = "abcabcbb"; // Expected: "abc", length 3 String s2 =
        "bbbbb";                // Expected: "b", length 1 String s3 =
        "pwwkew";               // Expected: "wke", length 3 String s4
        = "";                   // Expected: "", length 0

        System.out.println("Longest substring without repeating characters in s1: " +
lengthOfLongestSubstring(s1)); // Output: 3
        System.out.println("Longest substring without repeating characters in s2: " +
lengthOfLongestSubstring(s2)); // Output: 1
        System.out.println("Longest substring without repeating characters in s3: " +
lengthOfLongestSubstring(s3)); // Output: 3
        System.out.println("Longest substring without repeating characters in s4: " +
lengthOfLongestSubstring(s4)); // Output: 0
    }

    public static int lengthOfLongestSubstring(String s)
        { HashSet<Character> set = new HashSet<>();
        int maxLength = 0;
        int  start  =  0;
        int end = 0;

        while (end < s.length()) {
            char currentChar = s.charAt(end);
            if (!set.contains(currentChar)) {
                set.add(currentChar);
                maxLength = Math.max(maxLength, end - start + 1);
                end++;
            } else {
                set.remove(s.charAt(start));
                start++;
            }
        }

        return maxLength;
    }
}
```

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Arrays

1.) Find common elements between two arrays

```java
import java.util.HashSet;
import java.util.Set;

public class CommonElements {
    public static void main(String[] args)
        { int[] array1 = {1, 2, 3, 4, 5};
        int[] array2 = {4, 5, 6, 7, 8};

        Set<Integer> commonElements = findCommonElements(array1, array2);

        System.out.println("Common elements: " + commonElements);
    }

    public static Set<Integer> findCommonElements(int[] array1, int[] array2) {
        Set<Integer> set1 = new HashSet<>();
        Set<Integer> commonSet = new HashSet<>();

        // Add elements of the first array to the set
        for (int num : array1)
            { set1.add(num);
        }

        // Check for common elements in the second array
        for (int num : array2) {
            if (set1.contains(num))
                { commonSet.add(num);
            }
        }

        return commonSet;
    }
}
```

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Input: array1 = {1,2,3,4,5} and array2 = {4,5,6,7,8}

Output: Common elements: [4, 5]

2.) Find first and last element of Arraylist

```
import java.util.ArrayList;

public class Main {
     public static void main(String[] args)
     { ArrayList<String> arrayList = new
     ArrayList<>(); arrayList.add("Apple");
     arrayList.add("Banana");
     arrayList.add("Cherry");
     arrayList.add("Date");
     arrayList.add("Elderberry");

     if (!arrayList.isEmpty()) {
         String firstElement = arrayList.get(0);
         String lastElement = arrayList.get(arrayList.size() - 1);

         System.out.println("First element: " + firstElement);
         System.out.println("Last element: " + lastElement);
     } else {
         System.out.println("The ArrayList is empty.");
     }
 }
}
```

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Output:

First element: Apple

Last element: Elderberry

3.) Sort an array without using in-built method

```
public class Main {
      public static void main(String[] args)
      { int[] array = {5, 2, 9, 1, 6};

       selectionSort(array);

      System.out.println("Sorted array:");
         for (int num : array) {
            System.out.print(num + " ");
          }
   }

  public static void selectionSort(int[] array)
        { int n = array.length;
        for (int i = 0; i < n - 1; i++)
        { int minIndex = i;
           for (int j = i + 1; j < n; j++)
              { if (array[j] < array[minIndex])
              { minIndex = j;
            }
         }

        // Swap array[i] and array[minIndex]
        int temp = array[i];
        array[i] = array[minIndex];
        array[minIndex] = temp;
     }
   }
}
```

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Output:

Sorted array:

1 2 5 6 9

4.) Remove duplicates from an Array

```
import java.util.HashSet;
import java.util.Set;

public class Main {
    public static void main(String[] args)
        { int[] array = {5, 2, 9, 1, 6, 2, 5};

        int[] uniqueArray = removeDuplicates(array);

        System.out.println("Array with duplicates removed:");
        for (int num : uniqueArray) {
            System.out.print(num + " ");
        }
    }

    public static int[] removeDuplicates(int[] array)
        { Set<Integer> set = new HashSet<>();
        for (int num : array)
            { set.add(num);
        }

        int[] result = new int[set.size()];
        int i = 0;
        for (int num : set)
            { result[i++] = num;
        }

        return result;
    }
}
```

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Output:

Array with duplicates removed:

1 2 5 6 9

5.) Remove duplicates from an ArrayList

```
import java.util.ArrayList;
import java.util.HashSet;
import java.util.Set;

public class Main {
    public static void main(String[] args)
        { ArrayList<Integer> arrayList = new ArrayList<>();
        arrayList.add(5);
        arrayList.add(2);
        arrayList.add(9);
        arrayList.add(1);
        arrayList.add(6);
        arrayList.add(2);
        arrayList.add(5);

        ArrayList<Integer> uniqueList =
removeDuplicates(arrayList);

        System.out.println("ArrayList with duplicates
removed:");
        for (int num : uniqueList)
            { System.out.print(num + " ");
        }
    }

    public static ArrayList<Integer>
removeDuplicates(ArrayList<Integer> list) {
        Set<Integer> set = new HashSet<>(list);
        return new ArrayList<>(set);
    }
}
```

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Output:

ArrayList with duplicates removed:

1 2 5 6 9

6.) Find the missing number in an Array

```
public class Main {
    public static void main(String[] args) {
        int[] array = {1, 2, 4, 5, 6}; // Missing number is 3
        int missingNumber = findMissingNumber(array);
        System.out.println("The missing number is: " + missingNumber);
    }

    public static int findMissingNumber(int[] array) {
        int n = array.length + 1; // Since one number is missing, the length should be n+1
        int totalSum = n * (n + 1) / 2; // Sum of first n natural numbers

        int arraySum = 0;
        for (int num : array)
            { arraySum += num;
        }
        return totalSum - arraySum;
    }
}
```

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Output:

The missing number is: 3

7.) Find the largest and smallest element in an Array

```
public class Main {
    public static void main(String[] args)
        { int[] array = {5, 2, 9, 1, 6, 3};

        int[] result = findLargestAndSmallest(array);

        System.out.println("Smallest element: " + result[0]);
        System.out.println("Largest element: " + result[1]);
    }

    public static int[] findLargestAndSmallest(int[] array)
        { if (array == null || array.length == 0) {
            throw new IllegalArgumentException("Array must not be null or
empty");
        }

        int smallest = array[0];
        int largest = array[0];

        for (int num : array)
            { if (num < smallest)
            {
                smallest = num;
            }
            if (num > largest)
                { largest = num;
            }
        }
        return new int[]{smallest, largest};
    }
}
```

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Output:

Smallest element: 1

Largest element: 9

8.) Search element in an Array

```
 public class Main {
        public static void main(String[] args)
            { int[] array = {5, 2, 9, 1, 6, 3};
            int target = 6;

           int index = linearSearch(array, target);

           if (index != -1) {
             System.out.println("Element " + target + " found at index: " +
index);
        } else {
             System.out.println("Element " + target + " not found in the
array.");
         }
    }

    public static int linearSearch(int[] array, int target)
        { for (int i = 0; i < array.length; i++) {
            if (array[i] == target) {
                return i; // Element found, return index
            }
        }
        return -1; // Element not found
    }
}
```

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Output:

Element 6 found at index: 4

Element 10 not found in the array

9.) Array consists of integers and special characters,sum only integers

```
 public class Main {
       public static void main(String[] args) {
       String[] array = {"5", "2", "9", "a", "1", "6", "#", "3"};

       int sum = sumIntegers(array);

       System.out.println("Sum of integers in the array: " + sum);
}


 public static int sumIntegers(String[] array)
        { int sum = 0;
        for (String element : array)
        { try {
           int num = Integer.parseInt(element);
           sum += num;
        } catch (NumberFormatException e) {
           // Ignore non-integer elements
        }
     }
    return sum;
  }
}
```

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Output:

Sum of integers in the array: 26

10.) Find Minimum and Maximum from an Array

```
public class Main {
    public static void main(String[] args)
        { int[] array = {5, 2, 9, 1, 6, 3};

        // Find maximum and minimum int max
        = findMaximum(array); int min =
        findMinimum(array);

        // Print the results
        System.out.println("Minimum value in the array: " + min);
        System.out.println("Maximum value in the array: " + max);
    }

    public static int findMaximum(int[] array)
        { if (array.length == 0) {
            throw new IllegalArgumentException("Array must not be empty");
        }
        int max = array[0]; // Initialize max to the first element
        for (int i = 1; i < array.length; i++)
            { if (array[i] > max) {
                max = array[i]; // Update max if current element is larger
            }
        }
        return max;
    }

    public static int findMinimum(int[] array)
        { if (array.length == 0) {
            throw new IllegalArgumentException("Array must not be empty");
        }
        int min = array[0]; // Initialize min to the first element
        for (int i = 1; i < array.length; i++)
            { if (array[i] < min) {
                min = array[i]; // Update min if current element is smaller
            }
        }
        return min;     }
}
```

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Output:

Minimum value in the array: 1

Maximum value in the array: 9

11.) Java program to count Odd and Even number from given array

Input: {1,2,3,4,5,6,7,8,9}

```
public class Main {
      public static void main(String[] args)
         { int[] array = {1, 2, 3, 4, 5, 6, 7, 8,
         9};

         int[] count = countOddAndEven(array);

         System.out.println("Even numbers count: " + count[1]);
         System.out.println("Odd numbers count: " + count[0]);
}

public static int[] countOddAndEven(int[] array) {
         int[] count = new int[2]; // Index 0 for odd count, Index 1 for
even count

         for (int num : array)
            { if (num % 2 == 0) {
               count[1]++; // Increment even count
            } else {
              count[0]++; // Increment odd count
          }
        }
          return count;
     }
 }
```

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Output:

```
Even numbers count: 4

Odd numbers count: 5
```

12.) Java program – input array was

given [1,1,2,2,3,4,5,5,6,6],

Output – [3,4]

```
import java.util.HashMap;
import java.util.Map;
import java.util.ArrayList;
import java.util.List;

public class Main {
    public static void main(String[] args) {
        int[] array = {1, 1, 2, 2, 3, 4, 5, 5, 6, 6};
        List<Integer> result = findNonRepeatedElements(array);
        System.out.println("Non-repeated elements: " + result);
    }

    public static List<Integer> findNonRepeatedElements(int[]   @KushalParikh11
array) {

        // Step 1: Count occurrences of each element using a
HashMap
        Map<Integer, Integer> countMap = new HashMap<>();
        for (int num : array) {
            countMap.put(num, countMap.getOrDefault(num, 0) + 1);
        }

        // Step 2: Identify elements with count equal to 1 (non- repeated)
        List<Integer> nonRepeatedElements = new ArrayList<>();
        for (Map.Entry<Integer, Integer> entry :
countMap.entrySet()) {
            if (entry.getValue() == 1)
                { nonRepeatedElements.add(entry.getKey());
            }
        }
        return nonRepeatedElements;
    }
}
```

Output :

Non-repeated elements: [3, 4]

Java program to implement hashcode and equals

```
import java.util.Objects;

public class
Student
{ private int
id; private
String name;

// Constructor
public Student(int id, String name)
    { this.id = id;
    this.name = name;
}

// Getters and setters (omitted for brevity)

// hashCode method
@Override
public int hashCode() {
    return Objects.hash(id, name);
}

// equals method
@Override
public boolean equals(Object
    obj) { if (this == obj)
        return true;
    if (obj == null || getClass() !=
        obj.getClass()) return false;
    Student student = (Student) obj;
    return id == student.id && Objects.equals(name, student.name);
}

public static void main(String[] args) {
    // Creating objects of Student class Student
    student1 = new Student(1, "Alice"); Student
    student2 = new Student(2, "Bob"); Student
    student3 = new Student(1, "Alice");

    // Testing equals method
    System.out.println("student1.equals(student2): " +
student1.equals(student2)); // Output: false
    System.out.println("student1.equals(student3): " +
student1.equals(student3)); // Output: true

    // Testing hashCode method
    System.out.println("Hashcode of student1: " +
    student1.hashCode());  System.out.println("Hashcode of
```

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```
        student2:          "          +          student2.hashCode());
        System.out.println("Hashcode     of     student3:     "     +
        student3.hashCode());
    }
}
```

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